Bhagya Karunanayake

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ABOUT ME

Interested in hardware-software co-design to address real-world challenges, focusing on computer architecture, embedded systems, cybersecurity, networking, and AI/ML.

Education

University of Peradeniya, Sri Lanka Current GPA: 3.40 / 4.00
Bachelor of Science in Engineering (Hons) in Computer Engineering *2024 – 2028*

CP/Kot/Harangala Secondary School, Kothmale, Sri Lanka
Secondary Education

- G.C.E. Advanced Level Examination (Physical Science Stream) – 2022 | Island Rank: 978 / 32000+.
- Subjects: Combined Mathematics, Physics, and Chemistry – 3A's.

Projects

Panorama Generation from Overlapping Photos (Group) 2026
SIFT, Python, FLANN, OpenCV — GitHub

- Modular pipeline upgraded from ORB to SIFT + FLANN, a 10.4× increase in usable matches.
- Achieved sub-pixel geometric accuracy: RMSE reduced from 2.07 px to 0.54 px (3.8× better).
- Smart Fallback prevents extreme perspective distortion and Global Bundle Adjustment removes cumulative drift; compared overwrite, feathering, and Laplacian pyramid multi-band blending.

Fabric Defect Detection Using NNs (Group) 2026
Python, Jupyter Notebook — GitHub

- Classified high-resolution AITEX fabric images as defective or defect-free using neural networks.
- Automates textile quality control, replacing slow, subjective manual inspection prone to operator fatigue.

AI Sound Analyst & Health Manager for Industrial Assets (Group) 2026
TypeScript, Python, Jupyter Notebook, CSS, JavaScript, PL/pgSQL — GitHub

- PERA-SAM: predictive maintenance using FFT and MFCC acoustic analysis to detect wear before failure.
- AI Sound Doctor gives a real-time Health Score and Remaining Useful Life (RUL) estimate; technicians validate predictions (human-in-the-loop) to improve accuracy.
- Fleet web dashboard (Flask/Django); prototyped on fans, designed to scale to heavy machinery.

Panopticon: Real-Time AI Defense Against Voice Phishing (Group) 2026
C, C++, CUDA, Shell, Metal, Dart — Mobile Application — GitHub

- Fully on-device, multi-agent AI co-pilot for Android that transcribes live call audio with Whisper.cpp.
- Dual-LLM pipeline detects vishing, deepfake audio impersonation, and social engineering; a floating overlay shows the live transcript with a colour-coded risk level (Safe, Suspicious, Threat).
- Zero-egress design: all speech-to-text and LLM reasoning run on the phone, ensuring 100% data sovereignty.

Travel Bokka: Resilient Tourism & Cultural Navigator (Group) 2026
Dart, Python, C++, CMake, PL/pgSQL, Swift — Mobile Application — GitHub

- Offline-capable travel companion for Sri Lanka, built for AGENTRIX 2026 (Team 06, Tiramissu).
- Route Pivot Agent finds alternative routes via OSRM and generates localized negotiation scripts; Vision Agent gives cultural context from a photo of a monument.
- RAG knowledge base (customs, scam warnings, phonetic Sinhala); AI responses cached in Supabase.

Shadow AI Governance Detection (Individual) 2026
Python, Flask — GitHub

- Deterministic risk-scoring engine based on tool reputation and prompt sensitivity.
- Flags unapproved AI tools, PII leakage (email, credit card, SSN, IP via regex), and sensitive keywords; CSV/JSON reports and Flask dashboard.
- Verified with 100% precision and recall on synthetic datasets.

MIPS Processor Design (Individual) 2026

Verilog, C, Assembly, Shell — GitHub

- Designed a single-cycle 32-bit MIPS processor.
- MIPS: Microprocessor without Interlocked Pipeline Stages, a RISC-based architecture.

libtiny3d: 3D Software Renderer Library in C (Group) 2025

C, Makefile — GitHub

- Lightweight, portable pure-software 3D rendering library; no GPU required and minimal dependencies.
- Renders triangles, lines, and points through a modular, extensible pipeline.

cmd-lineshell: Command-Line Shell in ARM Assembly (Group) 2025

ARM Assembly — GitHub

- Custom Unix-like shell written entirely in ARM assembly, with a friendly error for unknown commands.
- Built-in commands: `hello`, `help`, `clear`, `exit`, `fac <n>`, `fib <n>`.

Experience

Secretary 2026 – Present

University of Peradeniya – ACES *Department of Computer Engineering, Sri Lanka*

Volunteer 2025

EngEX / Peraride *Sri Lanka*

- Volunteered at EngEX and with Peraride, an IoT-based automated bicycle sharing system.

Technical Skills

- **Hardware Programming:** Verilog, ARM Assembly
- **Programming Languages:** C, C++, Java, Python
- **Web Technologies:** HTML5, CSS, React, Node.js, Express
- **Automation & Workflow:** n8n, Make.com
- **Frameworks:** Flutter
- **Developer Tools:** Git, GitHub, Antigravity, VS Code, Colab, Linux
- **Data Modeling:** pandas, NumPy, Matplotlib
- **Cloud & DevOps:** Azure, Docker, Heroku

Achievements / Co-curricular Activities

Competitions 2026

University of Sri Jayewardenepura / University of Kelaniya / University of Ruhuna *Sri Lanka*

- Champions – AURORA AI Ideathon 2026, University of Sri Jayewardenepura.
- Semi-finalist – Codesplash Agentic AI Phase 2026, University of Kelaniya.
- 12th Place – AGENTRIX 2026, a 12-hour live hackathon at University of Ruhuna with 50+ teams.

Leadership and Associations 2018 – Present

University of Peradeniya / CP/Kot/Harangala Secondary School *Sri Lanka*

- Association of Computer Engineering Students (ACES) – Secretary (2026 – Present).
- Prefect of CP/Kot/Harangala Secondary School, Kothmale, Sri Lanka (2018).

Sports 2015 – 2025

University of Peradeniya / CP/Kot/Harangala Secondary School *Sri Lanka*

- Volleyball Team – University of Peradeniya (2025).
- College Volleyball Team – CP/Kot/Harangala Secondary School (2015 – 2020).
- Athletics – CP/Kot/Harangala Secondary School (2016 – 2020).

Referees

Prof. Roshan Ragel (roshanr@eng.pdn.ac.lk)

University of Peradeniya, Faculty of Engineering, Department of Computer Engineering *Sri Lanka*

Dr. Suneth Namal Karunarathna (namal@eng.pdn.ac.lk)

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